

10.18

a.

12.1515% of mothers have college education in this sample.

11.6868% of fathers have college education in this sample.

```
> mean(married$mothercoll)
[1] 0.1214953
> mean(married$fathercoll)
[1] 0.1168224
```

b.

corr(educ, mothercoll)=0.3595

corr(educ, fathercoll)=0.3985

corr(mothercoll, fathercoll)=0.3546

```
> cor(married[c("educ", "mothercoll", "fathercoll")])
          educ mothercoll fathercoll
educ      1.0000000  0.3594705  0.3984962
mothercoll 0.3594705  1.0000000  0.3545709
fathercoll 0.3984962  0.3545709  1.0000000
```

Parent-college dummies are moderately important for IV relevance. They may capture college exposure and attitudes toward higher education better than exact parental education years.

c.

95% CI for educ is [-0.001219763 ; 0.1532557]

```
> confint(mom_iv, "educ", level=0.95)
          2.5 %    97.5 %
educ -0.001219763 0.1532557
```

d.

1st stage

F = 63.56 >> 10; Pr(>F) = 1.455e-14 < 0.05

Linear hypothesis test:
mothercoll = 0

-> Reject H0 -> Mothercoll is a strong IV

Model 1: restricted model
Model 2: educ ~ exper + I(exper^2) + mothercoll

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.079094	0.303118	39.849	< 2e-16 ***
exper	0.056230	0.042101	1.336	0.182
I(exper^2)	-0.001956	0.001256	-1.557	0.120
mothercoll	2.517068	0.315713	7.973	1.46e-14 ***

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	424	1929.9	1	289.32	63.563	1.455e-14 ***

e.

wage equation

95% CI for educ is [0.02751845 ; 0.02751845]

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.2790819	0.3922213	-0.712	0.47714
educ	0.0878477	0.0307808	2.854	0.00453 **
exper	0.0426761	0.0132950	3.210	0.00143 **
I(exper^2)	-0.0008486	0.0003976	-2.135	0.03337 *

-> narrower than the one in (c)

```
> confint(momdad_iv, "educ", level=0.95)
          2.5 %    97.5 %
educ 0.02751845 0.1481769
```

f.

1st stage

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.890259	0.290251	40.965	< 2e-16 ***
exper	0.049149	0.040133	1.225	0.221
I(exper^2)	-0.001449	0.001199	-1.209	0.227
mothercoll	1.749947	0.322347	5.429	9.58e-08 ***
fathercoll	2.186612	0.329917	6.628	1.04e-10 ***

Pr(>F) = 2.2e-16 < 0.05 -> Reject H0

Linear hypothesis test:
mothercoll = 0
fathercoll = 0

-> Mothercoll and fathercoll are strong IVs.

Model 1: restricted model
Model 2: educ ~ exper + I(exper^2) + mothercoll + fathercoll

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	423	1748.3	2	470.88	56.963	< 2.2e-16 ***

g.

test the validity of the surplus instrument:

Sargan: $p\text{-value} = 0.626 > 0.05 \rightarrow$ do not reject $H_0 \rightarrow$ IVs are valid.

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	2	423	56.963	$<2e-16$ ***
Wu-Hausman	1	423	0.519	0.472
Sargan	1	NA	0.238	0.626

10.20

a.

```
y_A<- msft - riskfree
x_A<- mkt - riskfree
Intercept: 0.003250
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003250	0.006036	0.538	0.591
x_A	1.201840	0.122152	9.839	$<2e-16$ ***

x_A: 1.201840 $\rightarrow$ Beta > 1 meaning this stock is riskier than the market portfolio.

b.

```
1stage result:
beta_rank: 9.067e-04
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.903e-02	2.195e-03	-36.0	$<2e-16$ ***
rank	9.067e-04	2.104e-05	43.1	$<2e-16$ ***

t_value: 43.1 $\rightarrow$ the coefficient of rank is very significant

F-statistic: 1858 $\gg 10$

Multiple R-squared: 0.9126 Residual standard error: 0.01467 on 178 degrees of freedom
F-statistic: 1858 on 1 and 178 DF, p-value: $< 2.2e-16$

$\rightarrow$ Rank is a strong IV

c.

```
vhat_rank: -0.874599
Pr(>|t|): 0.0428 *
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003018	0.005984	0.504	0.6146
x_A	1.278318	0.126749	10.085	$<2e-16$ ***
vhat_rank	-0.874599	0.428626	-2.040	0.0428 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared: 0.3672, Adjusted R-squared: 0.36
F-statistic: 51.34 on 2 and 177 DF, p-value: $< 2.2e-16$

At level of 1%, $p\text{ value} = 0.04 > 0.01 \rightarrow$ do not reject H_0

Market return is exogenous at level of 1%

d.

Coefficients of x_A is 1.278318 > 1.201840

Coefficients of IV higher than coefficients of OLS

```
call:
ivreg(formula = y_A ~ x_A | rank, data = capm5)

Coefficients:
(Intercept)          x_A
      0.003018      1.278318
```

e.

1st stage result:

Test the joint significance of the IV:

$F = 951.26 \gg 10$

$\Pr(>F) = 2.2e-16 < 0.05 \rightarrow$ Reject H_0

$\rightarrow$ rank and pos are strong IVs.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.0804216	0.0022622	-35.55	<2e-16 ***
rank	0.0009819	0.0000400	24.55	<2e-16 ***
pos	-0.0092762	0.0042156	-2.20	0.0291 *

Linear hypothesis test:

rank = 0

pos = 0

Model 1: restricted model

Model 2: $x_A \sim \text{rank} + \text{pos}$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	179	0.43784				
2	177	0.03727	2	0.40057	951.26	< 2.2e-16 ***

f.

$\text{vhat_E} = -0.954918$

$\Pr(>|t|) = 0.0287 *$

At level of 1%, $p \text{ value} = 0.0287 > 0.01$

$\rightarrow$ do not reject H_0

Market return is exogenous at level of 1%

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003004	0.005972	0.503	0.6157
x_A	1.283118	0.126344	10.156	<2e-16 ***
vhat_E	-0.954918	0.433062	-2.205	0.0287 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared: 0.3696, Adjusted R-squared: 0.3625
F-statistic: 51.88 on 2 and 177 DF, p-value: < 2.2e-16

g.

$\#x_A: 1.283118 > 1.201840$ (a)

#Coefficients of IV higher than coefficients of OLS

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003004	0.006044	0.497	0.62
x_A	1.283118	0.127866	10.035	<2e-16 ***

h.

$\text{sargan_stat} = 0.5584634 < \text{critical_value} = 0.6702807$

Conclusion: the surplus IV is valid at the 5% level of significance.

```
> #NR^2
> N <- nrow(capm5)
> R2_aux <- summary(sargan_aux)$r.squared
> sargan_stat <- N*R2_aux
> #chi_squared test
> #df=L-B=1
> critical_value <- pchisq(0.95, df=1)
> if (sargan_stat > critical_value) {
+   cat("Reject H0: at least one of the IVs may be correlated with the error.")
+ } else {
+   cat("Fail to reject H0: no evidence that the surplus IV is invalid.")
+ }
Fail to reject H0: no evidence that the surplus IV is invalid.
```